Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 101606

Subject: Geotechnical Engineering - II

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer the following (Any seven question only): **[2 × 7 = 14]**

(a) The consolidation process is most significant in which type of soil?
(i) Sandy soil (ii) Clayey soil (iii) Gravelly soil (iv) Silty soil

(b) Which of the following is common test used to determine the consolidation properties of soil?
(i) Atterberg limit test (iii) Oedometer test
(ii) Standard proctor test (iv) Sieve analysis

(c) The Mohr-Coulomb failure criterion is used to describe-
(i) The permeability of soil (iii) The consolidation of soil
(ii) The relationship between stress and strain in soils (iv) The relationship between shear strength and normal stress

(d) What effect does an increase in moisture content have on the shear strength of clay soils?
(i) Increases shear strength (iii) Has no effect
(ii) Decreases shear strength (iv) Doubles the shear strength

(e) The most common type of slope failure is-
(i) Rockfall (iii) Rotational Slide
(ii) Earthfall (iv) Translational Slide

(f) Which of the following conditions is likely to lead to increased slope instability?
(i) Dry weather conditions (iii) Deep rooted vegetation
(ii) Heavy rainfall or snowmelt (iv) Low slope angles

(g) What is the importance of determining the groundwater level during soil exploration?
(i) It affects soil fertility (iii) It helps in plant selection
(ii) It influences the effective stress and shear strength of the soil (iv) It is irrelevant for construction purposes

(h) Geophysical methods in soil exploration include-
(i) Soil boring (iii) Cone penetration testing
(ii) Seismic & resistivity surveys (iv) Sieve analysis

(i) Which of the following is NOT typically a part of a soil exploration report?
(i) Soil stratigraphy (iii) Soil nutrient content
(ii) Groundwater levels (iv) Engineering recommendations

(j) Which of the following is NOT a method to calculate earth pressure?
(i) Rankine's theory (iii) Coulomb's theory
(ii) Mohr-Coulomb failure criterion (iv) Boussinesq's equation

Q.2 (a) What do you mean by soil consolidation? Also enlist primary factors affecting soil consolidation? **[7]**

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(b) For a soil undergoing consolidation. [7]
Initial void ratio = 0.7;
Final void ratio = 0.5;
Thickness of the clay layer = 5 m;
Specific weight of water = 9.81 kN/m³
Calculate:
(i) The change in void ratio (ii) The primary consolidation settlement

Q.3 (a) What is shear strength of soil and write down the factors affecting the shear strength of soil and their significance in geotechnical engineering. [7]

(b) A tri-axial test is conducted on a saturated clay sample. The following data is obtained: [7]
Cell pressure = 100 kPa and Deviator stress at failure = 250 kPa
Calculate:
(i) The effective normal stress at failure.
(ii) The shear strength parameters cohesion and angle of internal friction assuming a linear Mohr-Coulomb failure envelope.

Q.4 (a) What are the main factors affecting slope stability? Also explain the role of shear strength in slope stability? [7]

(b) A slope of height 12 m has a unit weight of soil of 18 kN/m³, cohesion of 15 kPa, and an internal friction angle of 28 degrees. An external load of 30 kN is applied at the top of the slope. Calculate the factor of safety. [7]

Q.5 (a) What methods are commonly used in soil exploration? Also explain what are the factors that influence the choice of soil exploration methods? [7]

(b) A soil layer is located at a depth of 5 m with a total stress of 100 kPa. The groundwater table is at 2 m depth, and the unit weight of the soil is 18 kN/m³. Calculate the effective stress at the depth of 5 m. [7]

Q.6 (a) Discuss the significance of consolidation in geotechnical engineering and its practical applications. [7]

(b) A saturated clay layer has a thickness of 4 m and a coefficient of consolidation (C_v) of 1×10^{-5} m²/s. A load of 100 kPa is applied on top of this layer. Assuming that the consolidation occurs under one-dimensional conditions, estimate the time required for 50% consolidation. [7]

Q.7 (a) Explain how laboratory tests are used to determine the shear strength of soil and describe common testing methods. [7]

(b) A direct shear test is performed on a sandy soil sample. The following data are recorded: [7]
- Normal stress applied = 150 kPa
- Shear force at failure = 90 kN
- Area of the shear box = 0.6 m²

Calculate:
1. The shear stress at failure.
2. The effective stress if the pore water pressure at failure is 20 kPa.
3. The cohesion and angle of internal friction using the Mohr-Coulomb failure criterion.

Q.8 (a) Describe the methods used for slope stability analysis and explain the wedge failure Swedish circle method in detail. [7]

(b) A slope of height H=10 m consists of saturated clay with the following properties: [7]
- Cohesion =25 kPa
- Angle of internal friction =15°

Assume the slope is inclined at an angle β=30°. Calculate the Factor of Safety (FOS) using the limit equilibrium method. For simplicity, assume a planar failure surface.

Q.9 (a) What is the purpose of soil instrumentation? Also enlist instruments that are commonly used in soil mechanics? [7]

(b) Write short notes on any two of the following: [7]
(i) Mechanism of Consolidation (ii) Mohr-Coulomb Failure Criterion
(iii) Difference between active and passive earth pressure
(iv) Standard Penetration Test (SPT)

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